CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

Some lectures, job postings, and FAQ are now available. See update log for update history and giving for donation. Contact info is given below.

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Membrane Builder

User Profile

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PDB Info | STEP 1 | STEP 2 | STEP 3 | STEP 4 | STEP 5 | STEP 6 | JOB ID: 3757172590

Oriented PDB:	step2_orient.pdb (view structure)
Component Input:	step4_lipid.inp
Component Output:	step4_lipid.out
Component Number:	step4_components.str
Component PDB:	step4_lipid.pdb (view structure)

download.tgz

Check lipid pentration

The protein surface penetration check finds the lipid tails that go beyond the protein surface, and the lipid ring penetration check detects the lipid tails that pass through the cyclic groups (e.g., cholesterol ring) in the simulation systems. Energy minimization can resolve many of these bad contacts, but one might need to visually check the following lipid molecules to ensure the following contacts are resolved. The user can regenerate the lipid bilayer if necessary.

Protein surface penetration:

```
No protein surface penetration is found.
```

Lipid ring penetration:

```
No lipid ring penetration is found.
```

Building Ion and Waterbox

Membrane components are generated. Due to time constrains, we first generate the lipid bilayer then generate ions and the water box. Click "Next Step" to generate ions and the water box.

Next Step:
Assemble Components

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